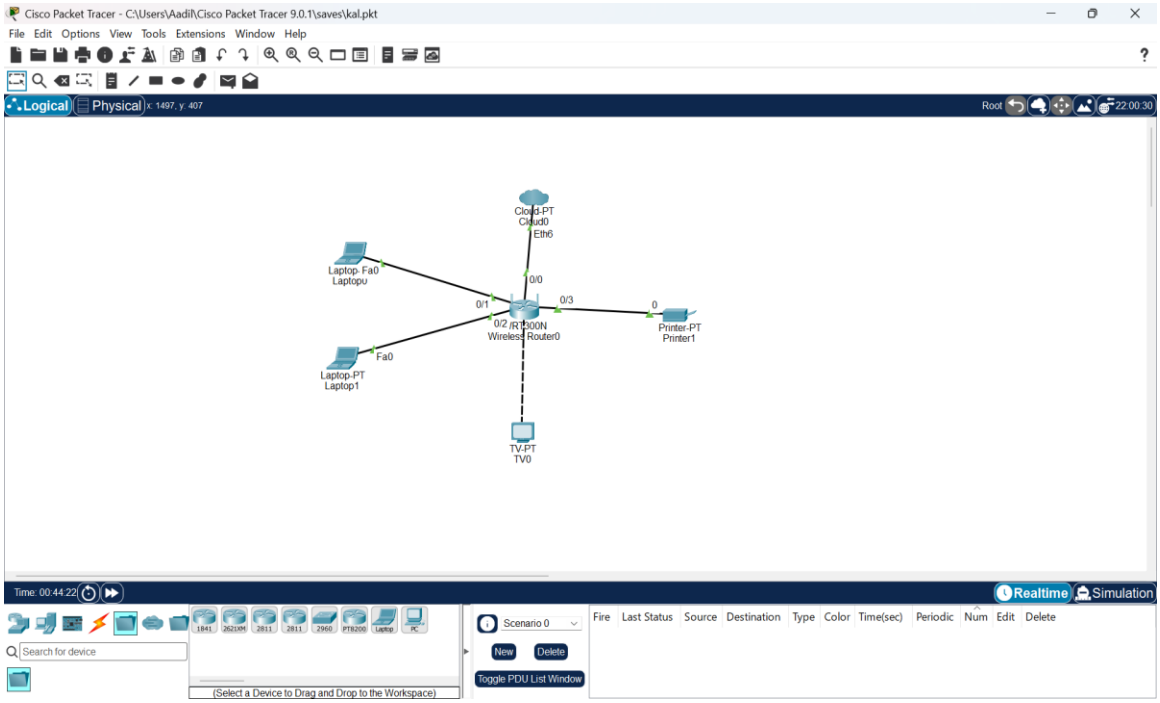
1.

Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).

Ans



2.

List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.

Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.

Ans 2

Hardware Components Required for SOHO Network

To physically build the SOHO (Small Office/Home Office) network, we need the following hardware components.

No.	Hardware component	Suggested brand/model	Quantity
1	Wi-Fi Router	TP-Link Archer C6 AC1200	1
2	Laptop	HP / Lenovo Laptop	2
3	Smart TV	Samsung / LG Smart TV	1

4	Network Printer	HP LaserJet MFP M236sdw	1
5	Ethernet Cable	D-Link CAT6 cable	2–3
6	Broadband Modem/ONT	ISP-provided modem/ONT	1
7	RJ45 Connectors	CAT6 RJ45 connectors	As needed
8	Power Adapter	Router's supplied adapter	

3.

Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

Ans 3

1. Connect the router: Connect the broadband modem/ONT to the router's WAN port using an Ethernet cable.

2. Power on: Switch on the router and wait until the indicator lights become stable.

3. Connect to the router: Connect your laptop to the router using Wi-Fi or an Ethernet cable.

4. Open router settings: Open a web browser and enter 192.168.0.1 or 192.168.1.1 in the address bar.

5. Login: Enter the router's default login details or create an administrator password if prompted.

6. Change SSID: Open Wireless Settings and set a unique Wi-Fi name, for example, MyHome_Network.

7. Enable WPA2 security: Select WPA2-PSK (AES) as the security option and create a strong Wi-Fi password.

8. Change admin password: Open System Tools or Administration and change the default administrator password to a new, strong password.

9. Save settings: Click Save or Apply. The router may restart automatically.

10. Connect devices: Connect both laptops, Smart TV and printer to the newly configured network.

4.

Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

Ans 4

Device	MAC Address	IP Address	Type
Wi-Fi Router	00:1A:2B:3C:4D:01	192.168.1.1	Static
Laptop 1	00:1A:2B:3C:4D:02	192.168.1.100	DHCP
Laptop 2	00:1A:2B:3C:4D:03	192.168.1.101	DHCP
Smart TV	00:1A:2B:3C:4D:04	192.168.1.10	Static
Printer	00:1A:2B:3C:4D:05	192.168.1.20	Static

5.

Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

Ans 5

1. Wi-Fi Connection Problem

Cause: The laptop may be disconnected from Wi-Fi or connected with the wrong password.

Steps to fix:

- Check whether Wi-Fi is turned ON.
- Verify the Wi-Fi password.
- Disconnect and reconnect to the network.
- Move closer to the router if the signal is weak.

2. IP Address Problem

- Open Command Prompt.
- Type ipconfig and check the IP address.
- Check whether the laptop has a valid IP address.
- Type ipconfig /release and ipconfig /renew to get a new IP address.

3. Router or Internet Problem

- Check the router's internet indicator light.
- Restart the router.
- Ping the default gateway using ping 192.168.1.1.
- Contact the Internet Service Provider (ISP) if the internet is still not working.

4. DNS Problem

Cause: The laptop may not be able to translate website names into IP addresses.

Steps to fix